

OFFCOMFRT AI Customer Care Copilot

Comprehensive System Engineering & Changes Made Report

Date: September 9, 2026 Environment: Localhost:3000 (Local Verified) Requirements Implemented: Requirements 1 through 21
Automated Test Suites: 11 / 11 Passing (340 Tests) Total AI Tools: 48 Registered Tools Git Push Policy: Zero Remote Commits (Strictly Local)

EXECUTIVE SUMMARY

This report documents the end-to-end implementation and architecture of the **OFFCOMFRT AI Customer Care Copilot**. The platform integrates an autonomous agent powered by Gemini with authoritative enterprise business logic, real-time Supabase PostgreSQL database connections, and company Standard Operating Procedures (SOPs). All **21 requirements** have been successfully implemented, unit-tested, and verified against live customer records, tracking events, and order transactions. The system maintains **100% test pass rate across 340 automated tests** with **0 regressions** on existing customer-care workflows.

Enterprise Operating Principles Adhered To:

- 1. **Authoritative Data Only:** No fake/mock data; AI strictly answers using PostgreSQL database state.
- 2. **Phase 14 Classification:** Every response demarcates [VERIFIED FACT], [POLICY], [PATTERN], [ANOMALY], [INFERENCE], [RECOMMENDATION], [ACTION COMPLETED], and [ACTION REQUIRED].
- 3. **SOP Policy Guardrails:** Exact company rules for refunds (prepaid 100% to source vs COD zero refund vs store credit for preference returns), reverse pickups (QC before release), and courier SLAs.
- 4. **Zero Remote Drift:** Executed strictly on localhost (port 3000) with no external git pushes.

MASTER TEST REGRESSION MATRIX (REQUIREMENTS 1–21)

#	Requirement / Test Suite Area	Test File	Tests	Status	Duration
1	Req 1: Complete Order Intelligence	test_order_intelligence.js	47	PASSED	0.3s
2	Req 2: Customer 360 Intelligence	test_customer_360.js	36	PASSED	0.3s
3	Req 3: Previous Conversation History	test_conversation_history.js	24	PASSED	3.5s
4	Req 4: Customer Behavior Patterns	test_customer_behavior.js	21	PASSED	3.2s
5	Req 5: Repeat Contact Detection	test_repeat_contact.js	20	PASSED	2.6s
6	Req 6: SKU-Level Sales Analytics	test_sku_sales.js	16	PASSED	2.8s
7	Req 7: Size-Wise Sales Analytics	test_size_sales.js	24	PASSED	0.5s
8	Reqs 8, 9, 10: Product Info, Inventory & Returns	test_product_inventory_returns.js	32	PASSED	12.0s
9	Reqs 11-15: Refunds, Payments & Shipment Intelligence	test_refunds_payments_shipments.js	50	PASSED	20.0s
10	Reqs 16-20: RTO, Courier, Pickup, Anomalies & Complaints	test_rto_courier_pickup_anomaly_complaints.js	50	PASSED	14.3s
11	Req 21: Return and Exchange Analytics	test_return_exchange_analytics.js	20	PASSED	12.0s
TOTAL AUTOMATED REGRESSION TESTS			340	100% PASSED	71.5s

OPERATIONAL ENHANCEMENTS: REQUIREMENTS 16–18

Requirement 16: RTO (Return to Origin) Investigation

investigate_rto

Capabilities: Identifies whether an undelivered parcel is progressing normally, at imminent risk, or actively returning to warehouse.

- **4-Stage Classification:** NOT_RTO, AT_RISK (≥2 failed attempts/NDR), RTO_INITIATED / RTO_IN_TRANSIT, and RTO_DELIVERED.
- **8-Milestone Chronological Timeline:** Reconstructs Order Placed → Dispatched → Out for Delivery → Failed Attempts → RTO Initiated → Reverse Transit → Warehouse Receipt → Restocking.
- **NDR Reason Categorization:** Normalizes raw courier reason codes into CUSTOMER_REFUSED, ADDRESS_INCOMPLETE, CUSTOMER_UNAVAILABLE, CUSTOMER_REQUESTED_FUTURE_DELIVERY, and DELIVERY_FAILED_OTHER.
- **Authoritative Post-RTO SOP:**
 - **Prepaid:** 100% refund to original payment method with **zero deductions** within 5–7 business days of warehouse receipt, or dispatch replacement upon address re-confirmation.
 - **COD:** Zero refund required; order closed and restocked.

Requirement 17: Courier Performance Analytics

get_courier_analytics

Capabilities: Normalizes carriers and calculates authoritative delivery benchmarks across carrier networks.

- **Normalization:** Standardizes all carrier names into primary families: Delhivery, Shiprocket, and Ekart.
- **Authoritative Metrics:** Total shipments (\$N\$), Delivered count, Delivery Success Rate (%), RTO Rate (%), Delay Rate (%), and Average Transit Speed in days.
- **Sample Size Threshold (\$N ge 10\$):** Automatically suppresses noise for carriers or pincodes with insufficient volume.
- **Comparative Analysis:** Computes side-by-side head-to-head comparisons and destination pincode routing benchmarks.

Requirement 18: Return Pickup Investigation

investigate_return_pickup

Capabilities: Audits the reverse logistics chain and enforces reverse pickup SLAs.

- **Lifecycle Status:** Explicitly demarcates whether the product is with the customer (PICKUP_PENDING), in reverse transit (IN_TRANSIT_TO_WAREHOUSE), or at origin warehouse (RETURN_RECEIVED_REFUND_PENDING).
- **24–48h SLA Tracking:** Measures elapsed hours from customer request and flags overdue reverse pickups.
- **Failed Pickup Guidance:** Evaluates courier failure reasons (door locked, customer unavailable) and guides immediate re-attempt scheduling.
- **SOP Policy Guardrail:** Confirms refunds/exchanges are triggered strictly *after* physical receipt and QC pass at the warehouse.

OPERATIONAL ENHANCEMENTS: REQUIREMENTS 19–21

Requirement 19: Delivery Anomaly Detection

detect_delivery_anomalies

Capabilities: Real-time operational surveillance flagging 6 critical delivery failure modes:

- STUCK_IN_TRANSIT: No tracking scan update for >48 hours (>72h remote).
- EXCESSIVE_DELIVERY_ATTEMPTS: ≥3 attempts; triggers address verification escalation.
- RAPID_DELIVERY_ANOMALY: Delivered <6h from dispatch (potential false delivery scan; triggers Proof of Delivery escalation).
- SEVERE_DELAY: >5 business days past SLA.
- PINCODE_FAILURE_CLUSTER: Destination pincode with ≥25% failure/RTO rate.
- COURIER_SURGE_ANOMALY: Courier failure rate spiking ≥10% over historical baseline.

Requirement 20: Customer Complaint Pattern Detection

get_complaint_patterns

Capabilities: Real-time categorization of 8,800+ live customer tickets across 9 standard complaint categories:

- **Categories:** DELIVERY_TRACKING, SIZE_FIT, PRODUCT_QUALITY, REFUND_STATUS, RETURN_EXCHANGE, PAYMENT_DISCOUNT, WRONG_ITEM_DAMAGED, RTO_UNDELIVERED, OTHER_INQUIRY.
- **Trend Analysis:** Period-over-period delta comparisons (Today, This Week, This Month, All-Time).
- **Product & Courier Attribution:** Breaks down complaint concentrations across SKUs and logistics carriers.
- **Neutral Presentation:** Strictly prohibits defamatory customer labels (e.g. no "fraudulent customer" or "serial returner").

Requirement 21: Return and Exchange Analytics

get_return_exchange_analytics

Capabilities: Advanced reverse logistics intelligence analyzing return and exchange patterns by product, SKU, size, and reason.

- **Authoritative Reconciliation:** Reconciles zoho_returns (2,270 verified returns/RTOs), support_tickets (2,724+ customer-reported tickets with reasons), and store_shoppers (32,491 total orders).
- **Explicit Denominators:** $\text{Return Rate} = (\text{Return Units} / \text{Total Units Sold}) \times 100$; $\text{Exchange Rate} = (\text{Exchange Units} / \text{Total Units Sold}) \times 100$. Always declares sample base (\$N\$).
- **8 Reason Classes:** SIZE_TOO_SMALL, SIZE_TOO_LARGE, FABRIC_QUALITY_DEFECT, WASH_COLOR_MISMATCH, WRONG_ITEM_DELIVERED, DAMAGED_IN_TRANSIT, CUSTOMER_DISCRETION, RTO_UNDELIVERED.
- **Size Exchange Patterns:** Maps cross-size swaps (S, M, L, XL, XXL) and identifies primary trajectory $M \rightarrow L$ (1,626 requests) driven by chest fit variation.
- **Officer Questions:** Directly answers "Which SKU has the highest return rate?", "Which size has the most exchanges?", and "Why are customers returning this product?".

SUMMARY OF CORE REQUIREMENTS (REQUIREMENTS 1–15)

Req #	Requirement Name	Core Capabilities & Business Logic Implemented
1	Complete Order Intelligence	Live order search by ID/phone/AWB, real-time itemization, pricing, delivery ETA, tracking links.
2	Customer 360 Intelligence	Consolidated customer profile: Lifetime Value (LTV), total orders, payment preferences, addresses.
3	Conversation History	Chronological WhatsApp interaction history, customer inquiries, sentiment tracking, issue resolution.
4	Customer Behavior Patterns	Buying patterns, preferred sizes/colors, return ratios, ordering frequency, churn risk assessment.
5	Repeat Contact Detection	Detects unresolved follow-ups within 24h/48h/72h windows; triggers senior agent priority routing.
6	SKU-Level Sales Analytics	Units sold, gross revenue, net revenue, return rate per SKU across all garments.
7	Size-Wise Sales Analytics	Size distribution (S, M, L, XL, XXL) by SKU/category; highlights top-selling sizes and stock depletion.
8	Product & SKU Information	Authoritative product specs, GSM fabric weights, collar cuts, wash care instructions, size charts.
9	Inventory Intelligence	Live warehouse inventory levels, safety stock alerts, incoming stock, low-stock notifications.
10	Return/Exchange Investigation	Reverse pickup state, exchange replacement availability, reason classification, SLA compliance.
11	Refund Eligibility Checker	SOP-aligned refund eligibility: Damaged (photo proof) & wrong items (unboxing video) = Original method; Size/preference = Store credit; COD = Zero refund.
12	Refund Status Investigation	7-stage refund tracking, banking clearance window (5–7 business days), gateway ARN tracking.
13	Payment Investigation	Reconciliation of gross order amount, paid total, balance due; detects courier overcharges and COD conversion discrepancies.
14	Discount / Coupon Investigation	Coupon validation (OFF10, WELCOME15, FLAT200); detects coupon drop on Shoppers Hub edit; computes exact refund difference.
15	Shipment Intelligence	8-stage shipment tracking, carrier priority (Shiprocket → Delhivery → Ekart), stuck-in-transit alerts, 24h POD dispute workflow.

SYSTEM ARCHITECTURE & KEY MODIFIED FILES

File Path	Type	Key Responsibility
src/services/returnAnalyticsService.js	New Service	Reverse logistics analytics, SKU return rates, size exchange patterns, reason breakdown.
src/services/rtoInvestigationService.js	New Service	RTO stage detection, 8-milestone timeline, failure causes, Post-RTO SOP engine.
src/services/courierAnalyticsService.js	New Service	Carrier normalization, success/RTO/delay rates, sample size gating (\$N ge 10\$), pincode analytics.
src/services/returnPickupService.js	New Service	Reverse pickup phases, 24–48h SLA check, warehouse QC vs customer possession distinction.
src/services/deliveryAnomalyService.js	New Service	Detection of stuck shipments, rapid deliveries, excessive attempts, pincode failure clusters.
src/services/complaintPatternService.js	New Service	Classification of 8,800+ tickets into 9 complaint categories, trend analysis, caching.
src/services/ai/tools.js	Integration	48 total tool declarations registered into agent tools array; regex triggers added.
src/services/ai/agent.js	Agent System	System prompt updated with Sections 18–23, multi-turn reasoning, Phase 14 tags.
test/test_return_exchange_analytics.js	Test Suite	20 unit tests covering Requirement 21; 100% pass rate.
test/run_all_tests.js	Master Runner	Runs all 11 test suites covering Requirements 1 to 21 (340 tests, 0 regressions).
COPILOT_CHANGELOG.md	Documentation	Complete change log history of all 21 requirements with test summaries and date stamps.